SHEAFIFICATION PRESERVES LIMIT PRESERVING TRANSFORMATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 667-691, following the statement at 658-665.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Fibre functor commutation with sheafification. equation-fibre-map-functors (same chapter, cited at ~674-675 and ~686-688). F^#_y is a sheaf and F^#_y = (F_y)^#.
- Locally limit preserving sheaves are limit preserving. lemma-sheaf-finite-presentation (same chapter, card L26299, cited at ~679-682). Glues local limit preservation over the fppf covering {V_j -> T}.
- Sheafification preserves limit preserving functors. lemma-sheafify-finite-presentation (same chapter, card L26298, cited at ~689-690). Concludes for the case y in G(T).

Source reference. Original definition 049J: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 287–287. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 049K: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 318–318. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 049P: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 597–597. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 049O: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 506–506. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $F, G : (Sch/S)^{opp}_{fppf} \to Sets$ be functors. If $a : F \to G$ is a transformation which is limit preserving, then the induced transformation of sheaves $F^\# \to G^\#$ is limit preserving.*

Proof. Suppose that T is a scheme and $y \in G^\#(T)$. We have to show the functor $F^\#_y : (Sch/T)^{opp}_{fppf} \to Sets$ constructed from $F^\# \to G^\#$ and y as in Definition 049J is limit preserving. By Equation (049K) we see that $F^\#_y$ is a sheaf. Choose an fppf covering $\{V_j \to T\}_{j \in J}$ such that $y|_{V_j}$ comes from an element $y_j \in F(V_j)$. Note that the restriction of $F^\#$ to $(Sch/V_j)_{fppf}$ is just $F^\#_{y_j}$. If we can show that $F^\#_{y_j}$ is limit preserving then Lemma 049P guarantees that $F^\#_y$ is limit preserving and we win. This reduces us to the case $y \in G(T)$.

Let $y \in G(T)$. In this case we claim that $F^\#_y = (F_y)^\#$. This follows from Equation (049K). Thus this case follows from Lemma 049O. □